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Principles of Computer Vision

Module: IN2133 **Date:** Thursday 27th February, 2025
Examiner: Prof. Dr.-Ing. Darius Burschka **Exam:** Endterm

First name: Qirui **Registration number, UIDs:** 03780247, E0111 / –
Last name: Wang **Last update:** Wednesday 12th March, 2025

Σ (Endterm)	Grade (Endterm)	Grade intervall
49.0	3.3	[48.0; 57.0)

Notes:

- Please make sure that the total amount of credits stated above is correct.

Corrections:

The table below lists all corrections (image recognition and complaints during review) that are already considered in the calculation of your grade. If a problem or subproblem is listed multiple times, the correction with the highest number (column "Correction") takes precedence.

Problem	Correction	credits	Annotations



	0	3	7	8	0	2	4	7
1								
2								
3								
4								
5								
6								
7								
8								
9								
0								

Registration number

Qinui, Wang

Signature

Note:

- Cross your Registration number(with leading zero). It will be evaluated automatically.
- Sign in the corresponding signature field.

Principles of Computer Vision

Exam: IN2133 / Endterm

Date: Thursday 27th February, 2025

Examiner: Prof. Dr.-Ing. Darius Burschka

Time: 11:00 – 12:00

Working instructions

- This exam consists of **12 pages** with a total of **2 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 135 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one **non-programmable pocket calculator**
 - one **analog dictionary** English ↔ native language
- Subproblems marked by * can be solved without results of previous subproblems.
- **Answers are only accepted if the solution approach is documented.** Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from _____ to _____ / Early submission at _____



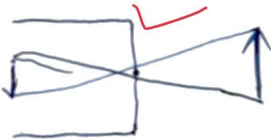
Problem 1 Image Formation (78 credits)

A camera with the perspective camera model is used to acquire images of the environment. A lens with the focal length $f_L = 8\text{mm}$ is used in front of a 512×512 pixels CCD chip with a pixel-size $p_x = 7\mu\text{m}$.

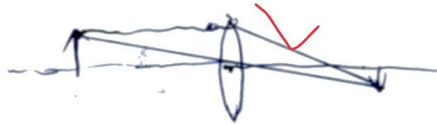
4.0 a)* Lens system

- Sketch briefly the difference in the collected light between a primitive pinhole camera and a camera with a lens mounted in front of the CCD chip;
- List in a bulleted list, what is improved and what additional problems get caused by adding a lens to the system (name 1 positive and 3 negative effects);
- Explain with additional drawing, what causes a limited distance range in front of the camera that creates a sharp image and name this effect;
- How can this range be extended? Add to the drawing a solution that will extend this distance range and provide the correct name for this solution.

Pinhole



lens



lens: Adv It can ~~collect~~ converge light and collect more lightness.

~~Dis~~ Negative Effects. ①. It will introduce Errors.

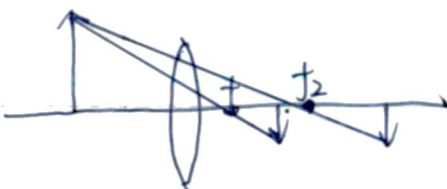
②. It will lead to distortion especially radial distortion.



This is caused by depth of field.

when

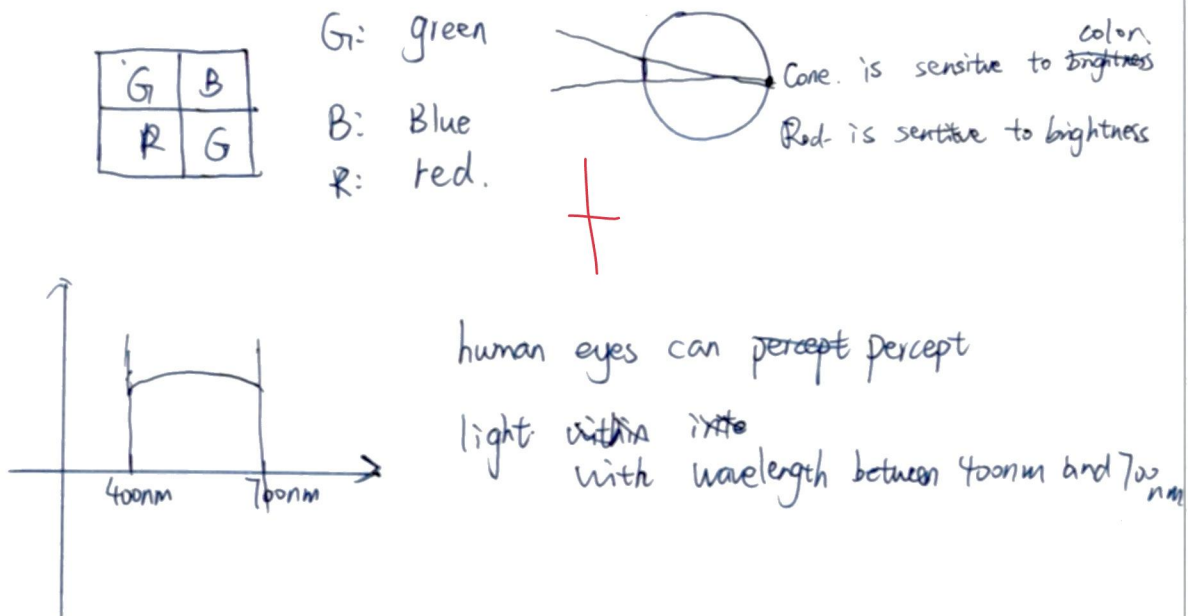
We can increase the focal length.





2.0b)* A human eye senses the incoming light using various photosensitive elements on the retina. The eye is capable of sensing a *visible spectrum* using this structure.

- Explain and draw, how the eye manages to sense the *visible spectrum* with a limited number of sensing elements in each sensed pixel naming the corresponding elements
- Sketch the spectral sensitivity of elements responsible for color perception
- Explain "metamerism" sketching examples of contributing spectral distributions of the incoming light.



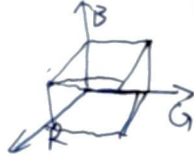
Matamerism means

Two different spectral distributed light can lead to the same response to human eyes

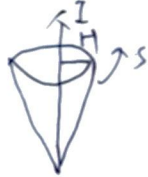


- 10.0 c)* Draw the color spaces used for image synthesis on screens and for image analysis and explain, why not the same color space is used? What is the advantage of the color space used for analysis? Explain the term *saturation* and *hue* in a separate drawing and draw spectral distributions for fully saturated green and 60% saturated green.

In Image synthesis we use RGB, which is a cube in 3D space



In Image analysis we use HSI which is a cone in 3D space.

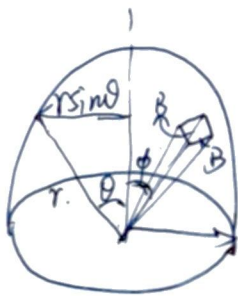


Because human eyes are more sensitive to saturation, ~~hue~~ so HSI is better for analysis.



The distance between surface of cone and axis is Hue. And Saturation is the angle spinned around the axis.

- 10.0 d)* How can you express the solid angle $d\omega$ that is used to integrate the incoming light energy on a patch using the angles of the hemisphere. Draw the parameterization of the solid angle on the hemisphere and derive the equation for $d\omega$:



$$ds = r^2 \sin\theta \, d\theta \, d\phi$$

$$= r \cdot \sin\theta \, d\theta \cdot r \, d\phi$$

$$= r^2 \sin\theta \, d\theta \, d\phi$$

$$d\omega = \frac{ds}{r^2} = \sin\theta \, d\theta \, d\phi$$



- 5.0 e)* Consider a rectangular uniformly shining surface with diagonal corners at $(w, d/2, 0)^T, (w, d/2, h)^T, (w, -d/2, h)^T, (w, -d/2, 0)^T$ as depicted in Figure below. The total radiance of the entire rectangle is I .

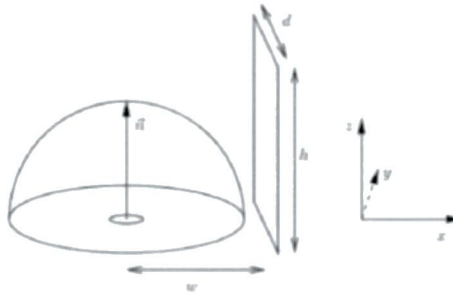


Figure 1: Problem setup.

The intensity E illuminating a point in the middle of the half-sphere H_z can be calculated as

$$E = \int_{D \in H_z} L(\omega) \cos \theta d\omega$$

- Which effect is reflected in the cosine expression estimating E above?
- Explain, how the expression $d\omega$ can be calculated for a half-sphere with the radius $r=1$ using the azimuth angle ϕ and elevation angle Θ ?
- What is the intensity arriving at the center of the half-sphere $(0,0,0)$ with normal vector $(0,0,1)$ (see Figure) from a point $p = (x, y, w)^T$ on the rectangle? Try to express the function in the integral above a function of (w, y, z, dz, dy) .
- How does the full integral including integration boundaries look for this example?

$$(0 \ 0 \ 1) \cdot \begin{pmatrix} x \\ y \\ w \end{pmatrix}$$

$$dw = \sin \theta d\theta d\phi$$

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ w \end{pmatrix}$$

I can be obtained via formula

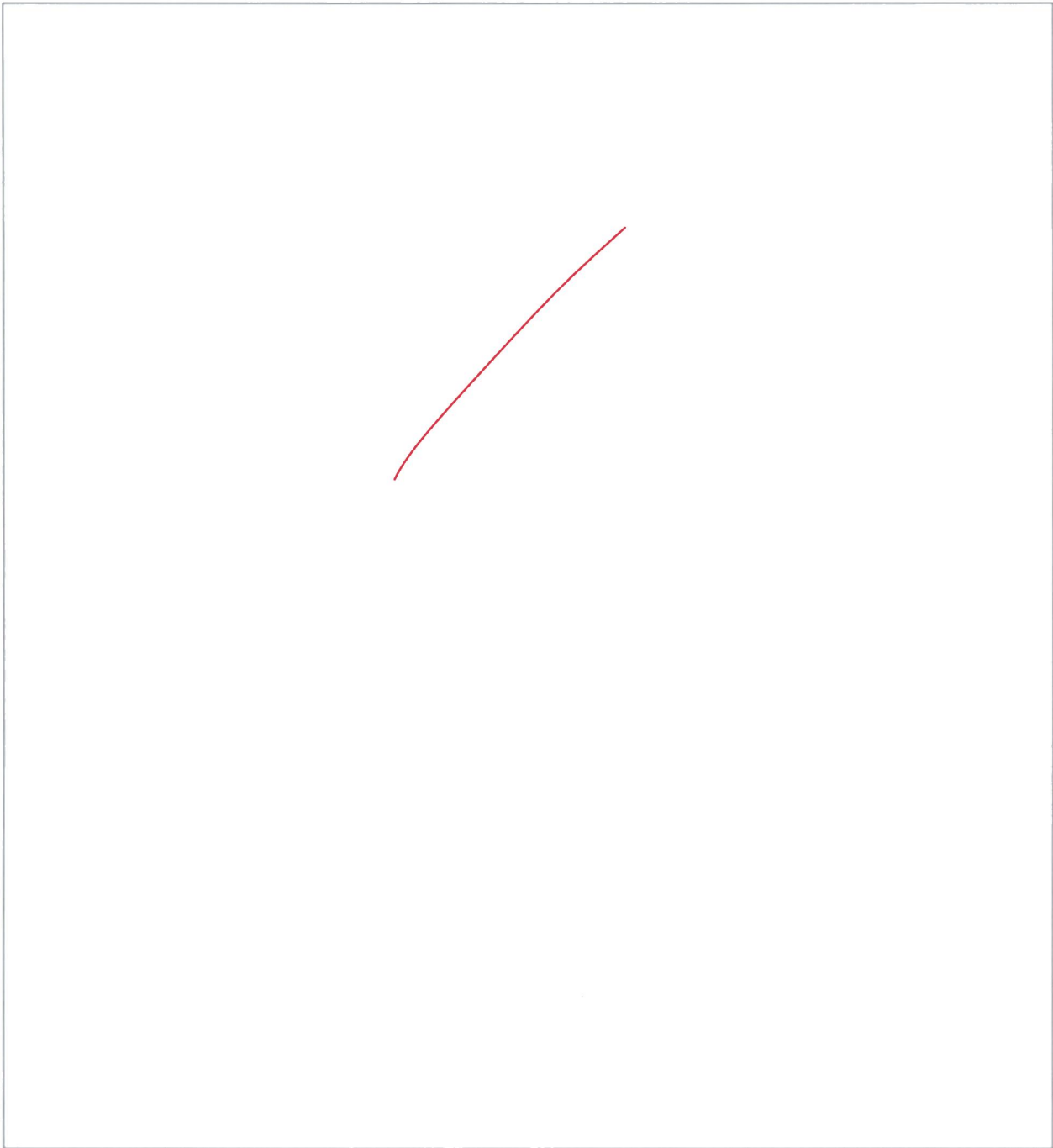
$$I = \frac{(p \cdot q + 1)^T \cdot (p_s \cdot q_s + 1)}{\|p \cdot q + 1\| \cdot \|p_s \cdot q_s + 1\|}$$

$$= \frac{(0 \ 0 \ 1)^T \cdot (x \ y \ w) \cdot (x \ y \ w) \cdot (0 \ 0 \ 1)}{\| (0 \ 0 \ 1)^T \cdot (x \ y \ w) \| \cdot \| (x \ y \ w) \cdot (0 \ 0 \ 1) \|}$$

$$= \frac{w}{x+y+w}$$

$$\text{So } E = \int \frac{w}{x+y+w} \cos \theta \cdot dw$$





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Problem 2 Reconstruction (57 credits)

7.0a)* A binocular camera system is used to measure depth of objects in range $z = [1 - 3]$ with an accuracy better than $\Delta z = 0.01m$. The cameras have both identical focal lengths $f/p_x = 800pixels$ and the matching points in the images are detected with an accuracy of 0.5pixels

- How can you calculate z (along optical axis) for an ideal binocular system from matched observations?
- Draw the curve relating the z -distance of the reconstructed points to the value estimated from the images.
- How can you derive the depth variation relative to image signal variation of the binocular system?
- Based on above equation, what is the baseline B of the system that needs to be chosen to get the required accuracy?

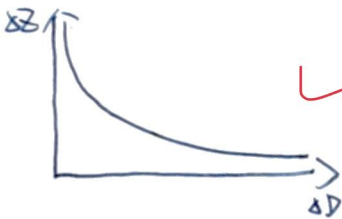


$$z = \frac{Bf}{Dp_x}$$

We can use formular

$$z = \frac{Bf}{Dp_x} \text{ to calculate } z.$$

In which B is baseline, f/p_x is focal length and D is disparity which can be $x_l - x_r$.



$$z = \frac{Bf}{Dp_x}$$

we can differentiate

$$\frac{dz}{dz} = \frac{Bf}{z^2} \rightarrow \frac{\Delta z}{z^2} = \frac{z^2}{Bf} \Delta D$$

Based on the equation we can obtain:

$$0.01m = \frac{4 \cdot 0.5 \text{ pixel}}{B \cdot 800 \text{ pixel}} \Rightarrow B = 0.25m$$

The Baseline of the system should be 0.25m.

$$0.01 = \frac{z^2}{B \cdot 800 \text{ pixel} \cdot 0.5 \text{ pixel}}$$

$$0.01m = \frac{4}{B \cdot 800 \cdot p_x}$$

$$0.01 = \frac{z^2}{800 B}$$

80

$$800 = 2 \quad B = \frac{z^2}{8}$$



- 0.0 b)* Explain using equations, how the inner structure of the Essential Matrix can be derived. What does this matrix relate to each other?

$$\vec{P} \times \vec{T} = \vec{n}$$

$$\begin{cases} \vec{P} = z\vec{p} + \vec{T} \\ z\vec{p} = \vec{P} - \vec{T} \end{cases} \quad \vec{p} \cdot \vec{n} = 0$$

we know that:

$$\vec{P} = zR\vec{p} + \vec{T}$$

Therefore:

$$\begin{aligned} zR\vec{p} + \vec{T} \cdot \vec{p} \cdot \vec{n} &= \vec{P} \cdot \vec{T} \\ &= (zR\vec{p} + \vec{T} - \vec{T}) \cdot (\vec{P} - \vec{T} \times \vec{P}) \\ &= (zR\vec{p}) \cdot \vec{T} \times \vec{P} = 0 \\ &= \frac{1}{2} z \vec{p} \cdot \text{skew}(\vec{T}) \cdot \vec{p} \\ E &= R \cdot \text{skew}(\vec{T}) \end{aligned}$$

- 1.0 c)* Show, why verged binocular camera system increases the spatial resolution of the stereo system compared to a co-planar system setup with similar field of view. Use the sensitivity equation dd/dz discussed in the lecture.

$$\frac{dd}{dz} = \frac{B}{f} \frac{1}{z^2}$$

Therefore: $\Delta Z = \frac{z^2}{Bf} \Delta d$

B: Baseline
f: focal length
z: Depth

Therefore. If we want to increase the spatial resolution, we can decrease B, f or increase Δd .

As in the problem stated, verged binocular camera system ~~decrease~~ exactly ~~increase~~ the Baseline. so the spatial resolution is increased.



4.0 d)*

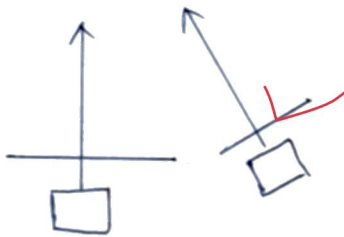
- What are epipolar lines and how are they defined?
- Sketch the camera arrangement (position and orientation of image planes, focal points and optical axes) for the following 2 epipolar geometries.
- Justify your choice by explaining where the epipoles lie and how they can be found.



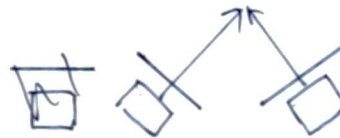
Epipolar lines

Epipolar lines is the intersection between epipolar plane and image plane.

First case:



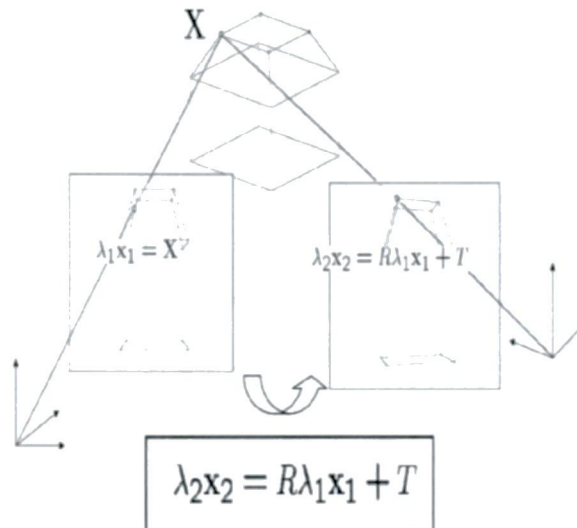
Second Case.



Epipoles lines connect the focal points and epipole. which is defined as the intersection between baseline and image plane.

Since in the first case, one is horizontal and one is slope, so it should be one camera ~~not~~ exactly forward and one slope. In second two epipolar line are slope so it should be the case.





6.0 e)* A point observed in two images

- Derive the equation that estimates the distances λ_1, λ_2 along the rays of observation in both cameras.
- Explain, while a pseudo-inverse is used in the solution and what does it mean for non-ideal calibration of the camera system.

~~$\lambda_2 x_2 = R \lambda_1 x_1 + T$~~

Since ~~$\lambda_2 x_2 = R \lambda_1 x_1 + T$~~ $\lambda_2 x_2 = R \lambda_1 x_1 + T$.

$$\lambda_2 x_2 - R \lambda_1 x_1 = T$$

we can obtain. $[\lambda_1 \lambda_2] \cdot \begin{bmatrix} -R x_1 \\ x_2 \end{bmatrix} = T$

It can be stated as $[\lambda_1 \lambda_2] \cdot s = T$.

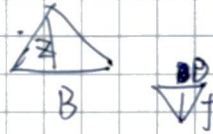
If ~~the~~ s is not square, so we have to use pseudo-inverse.

For non-ideal calibration of the camera system we should also take into account distortion coefficients $k_1, \dots, k_n$.



Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

$$L_2 X_2 =$$



$$\frac{Z}{B} = \frac{1}{D}$$

$$Z =$$

$$ZD = B$$

$$Z = \frac{B}{D}$$

10

$$P = L_1 X_1$$

$$X_1 = L_1 X_1$$

$$X_2 = L_2 X_2$$

$$X_2 = R \cdot X_1 + T$$

$$\cancel{X_1} = \cancel{R \cdot X_1} + T$$

$$L_2 X_2 = R \cdot L_1 X_1 + T$$

$$L_2 X_2 - R L_1 X_1 = T$$

$$\begin{bmatrix} L_2 & -R L_1 \end{bmatrix} \begin{bmatrix} X_2 \\ X_1 \end{bmatrix} = T$$

$$\begin{bmatrix} L_1 & L_2 \end{bmatrix} \oplus \begin{bmatrix} R X_1 \\ X_2 \end{bmatrix} = T$$

$$\begin{bmatrix} -R L_1 & L_2 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = T \quad \cdot \quad \begin{bmatrix} L_1 & L_2 \end{bmatrix} \cdot S = T$$





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Handwritten notes on lined paper, likely a student's work, showing a sequence of mathematical expressions and symbols. The text is written in a cursive, somewhat slanted style. The visible content includes:

- Top line: $\frac{1}{2} \cdot R \cdot P \cdot \text{Skent} \cdot \frac{1}{2}$
- Second line: $\frac{1}{2} \cdot R \cdot P \cdot \text{Skent} \cdot \frac{1}{2}$
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